

6. Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{Var}(e_i|x_i, \text{FEMALE} = 0) = \sigma_M^2$ and $\text{Var}(e_i|x_i, \text{FEMALE} = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

We test

$$H_0: \sigma_M^2 = \sigma_F^2, H_1: \sigma_M^2 \neq \sigma_F^2.$$

The estimated variances are:

$$\hat{\sigma}_M^2 = \frac{97161.9174}{573} \approx 169.57, \hat{\sigma}_F^2 = 12.024^2 \approx 144.58$$

The F-statistic is:

$$F = \frac{169.57}{144.58} \approx 1.173$$

At the 5% significance level with large degrees of freedom, the rejection region is approximately:

$$F > 1.196781 \text{ or } F < 0.837669$$

Since $F = 1.173$ is within this interval, we fail to reject H_0 . There is no evidence of different error variances between males and females.

b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages.

Suppose $\text{Var}(e_i|x_i, \text{MARRIED} = 0) = \sigma_{SINGLE}^2$ and $\text{Var}(e_i|x_i, \text{MARRIED} = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add FEMALE to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i$$

Using $N = 400$ observations on single individuals, OLS estimation of the above equation yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

We test:

$$H_0: \sigma_{SINGLE}^2 = \sigma_{MARRIED}^2, H_1: \sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$$

The estimated variances are:

$$\hat{\sigma}_{SINGLE}^2 = \frac{56231.0382}{(400 - 4 - 1)} \approx 142.3571$$

$$\hat{\sigma}_{MARRIED}^2 = \frac{100703.0471}{(1000 - 400 - 4 - 1)} \approx 169.2488$$

The F-statistic is:

$$F = \frac{169.25}{142.36} \approx 1.1890$$

At the 5% significance level, the rejection region is:

$$F > F_{0.95}(595, 395) \approx 1.164705$$

Since $F = 1.1890 > 1.1647$, we reject H_0 . There is evidence that married individuals have higher wage variability.

c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

The NR^2 test statistic is 59.03. Under the null hypothesis of homoskedasticity:

$$NR^2 \sim \chi^2(4)$$

At the 5% significance level, the critical value is:

$$\chi_{0.95}^2(4) \approx 9.487729$$

Since $59.03 > 9.49$, we reject the null hypothesis. There is strong evidence of heteroskedasticity.

However, this test does not provide direct evidence regarding whether the variance differs between married and unmarried individuals. The NR^2 test examines whether the error variance is related to all explanatory variables, not just marital status.

d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i$$

The White test statistic follows a chi-square distribution with degrees of freedom equal to the number of regressors in the auxiliary regression excluding the intercept.

The auxiliary regression includes:

- 4 original regressors
- 2 squared terms
- 6 interaction terms

Thus,

$$df = 4 + 4 + 6 - 2 = 12$$

At the 5% significance level, the critical value is:

$$\chi^2_{0.95}(14) \approx 21.02607$$

Since $78.82 > 21.03$, we reject the null hypothesis of homoskedasticity. There is strong evidence of heteroskedasticity.

e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{cccccc} \widehat{WAGE} = -17.77 + 2.5EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE \\ (se) \quad (2.36) \quad (0.14) \quad (0.031) \quad (1.05) \quad (0.81) \\ (robse) \quad (2.50) \quad (0.16) \quad (0.029) \quad (0.84) \quad (0.80) \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

	<i>EDUC</i>	<i>EXPER</i>	<i>METRO</i>	<i>FEMALE</i>
t_{OLS}	$\frac{2.5}{0.14} \approx 17.86^{***}$	$\frac{0.23}{0.031} \approx 7.42^{***}$	$\frac{3.23}{1.05} \approx 3.08^{***}$	$\frac{-4.20}{0.80} \approx -5.19^{***}$
t_{robust}	$\frac{2.5}{0.16} = 15.63^{***}$	$\frac{0.23}{0.029} \approx 7.93^{***}$	$\frac{3.23}{0.84} \approx 3.85^{***}$	$\frac{-4.20}{0.81} \approx -5.25^{***}$

The width of confidence intervals depends on the standard errors.

Comparing usual and robust standard errors:

- EDUC and EXPER have larger robust standard errors, so their confidence intervals become wider.
- METRO, FEMALE, and the intercept have smaller robust standard errors, so their confidence intervals become narrower.

To assign significance levels, t-statistics are computed as coefficient divided by standard error.

Comparing OLS and robust standard errors, all variables remain highly significant under both specifications.

The significance indicators (*, **, ***) are identical across OLS and robust standard errors.

Therefore, there is no inconsistency in the results, and the statistical inference is robust to heteroskedasticity.

f. If we add *MARRIED* to the model in part (b), we find that its *t* – value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

The t-value of approximately 1.0 for MARRIED indicates that the coefficient is not statistically significant when using heteroskedasticity-robust standard errors.

However, this does not conflict with the result in part (b). The heteroskedasticity test in part (b) is a

joint test examining whether the variance of the errors depends on any of the explanatory variables, whereas the t-test for MARRIED only evaluates the significance of a single coefficient in the mean equation.

Therefore, the two results are compatible, as they address different hypotheses.